SECTION A

Answer all items in this section

Item 1

Cancer has become an increasingly serious health challenge in Uganda. A privately owned hospital intends to establish a specialized cancer treatment unit. As a cost effective option, the hospital plans to acquire an efficient teletherapy machine that uses radioactive cobalt $^{60}_{27}C_o$, as a source of gamma radiation for destroying cancerous tissues.

The isotope $^{60}_{27}C_o$, undergoes radioactive decay to form nickel, emitting a beta particle and gamma rays in the process. Before approving the purchase a group of concerned staff recommended that the hospital management first gain a clear understanding of atomic stability using the binding energy per nucleon graph, formulate the nuclear decay equation, determine the rate of radioactive disintegration and estimate the energy that can be released during the decay process.

Available reference information shows that an efficient teletherapy machine utilizes 30g of $^{60}_{27}C_o$ with a disintegration rate not exceeding $9 \times 10^{14} Bq$, while producing atleast 5.5MeV of energy.

To ensure that the most appropriate conclusions are reached, the project manager was advised to seek assistance from an advanced level S. 6 physics student. The student is expected to describe the structure and working principle of a device that employs particles possessing properties similar to those of beta particles to generate radiations used in medical imaging for detecting bone fractures in cancer patients who frequently complain of bone pain.

Support material

Half-life of $^{60}_{27}C_o$ is 5.27 years.

Mass of $^{60}_{27}C_o$ = 59.93381U.

Mass of Ni = 59.93079U.

Mass of an electron = 0.0005U

1U = 931Mev.

Task

You have been selected for help

(a) Analyse the given data to address the concerns raised by health workers.

(b) Explain how the device suggested can work to serve the desired purpose.

Item 2

Uganda's growing involvement in space science was demonstrated by the successful launch of the Satellite Pearl African sat -1 in November 2022. To strengthen its space programme, plans are underway to prepare for another satellite launch before 2030. As part of these preparations, a satellite of mass 250kg is to be placed in a parking orbit from the Mpoma East ground station in

Turn over

Mukono. The mission team requires information on the height of the parking orbit, the mechanical energy of the satellite and the implications of reducing the satellite's mechanical energy while in the orbit. This information is needed to guide operational and safety considerations.

To obtain the required calculations urgently, the project supervisor contacted a space Engineer who was at his residence near the station. The Engineer was instructed to report to the station within five minutes.

While travelling he was accompanied by an S. 6 Physics student on a study visit and requested the student's assistance with the required analysis.
The Engineer explained that the vehicle was parked on a straight road leading directly to the station. The engine would supply a power of 3.0KW, producing a driving force of 1.8KN. The vehicle was expected to accelerate uniformly over the first 0.6km of the journey and then decelerate uniformly to rest over the remaining 0.2km before reaching the station. The student was asked to determine whether they could arrive at the station within the required time.

In addition, the Engineer requested the student to explain how locally available materials could be used to construct a device capable of generating lift and carrying amounted camera for capturing aerial photographs at the station.

Support material:
Coefficient of friction on the used road = 0.025.
Mass of the car = 1 metric tonne.
Acceleration due to gravity = $9.81ms^{-2}$
Radius of the Earth, Re = $6.4 \times 10^{6}ms^{-2}$
The packing orbit is 300km above the Earth.

Task
You have been selected to move with the Engineer;
(a) Analyse the data to provide the necessary assistance.
(b) Explain how the device suggested can be made and able to serve the purpose.

SECTION B

Part 1
Answer one item from this part.

Item 3

Musical instruments like guitars are commonly used in Uganda for performance, learning and entertainment, Samuel, a well-known business man in town, deals in selling these instruments. Recently, several customers have raised concerns about the heating effect of guitar strings.

To investigate the issue, Samuel asked his senior six physics student, who was helping at the shop over the weekend, to invite a Luther to examine the guitars. The Luther carried out a test on a string fixed at both ends, with a length of 40cm and a mass of 0.0012kg. He adjusted the string so that it produced a third harmonic of a frequency of 750Hz. He then asked the students to calculate the tension developed in the string in order to verify the customers' concerns and explain the heating effect observed in the strings.

In addition, the Luther instructed the student to derive the equation describing waves on a stretched string, showing how displacement relates to amplitude, so that the properties of such strings can be analysed in the feature.

Meanwhile, Samuel stepped onto the balcony and observed the day time moon using a refracting instrument set under normal adjustment, which had been given to him as a gift. He was fascinated by what he saw. At the same time, he was drinking cold soda from a glass bottle.

After a short while, he noticed droplets of liquid forming on the outer surface of the bottle, which made him wonder whether the glass was porous. Curious, he called the student to explain the observation.

Task

(a) Help the Luther to address the customers' concerns and obtain the desired expression for future use.

(b) Explain to Samuel what happened on the outside of the soda bottle and describe for him the structure and operation of the device he received as a gift.

Item 4

A secondary school experienced complaints from students about excessive heating in one of the classrooms. The students suggested that the problem could be caused either by sunlight entering through the door whenever it was left open or by the properties of the recently installed windows.

To investigate the matter, the headteacher organized a meeting involving the students, an advanced level physics teacher, several S. 6 students and the supplier of the windows. The supplier maintained that the windows met the required standards. He explained that each window consisted of a double-glazed unit made of two glass panes each 6m thick, separated by a 12mm layer of air. The window had a total cross sectional area of $30m^2$ and received 65% of the incident solar radiation from sun of solar constant $1400Wm^{-2}$. He further stated that the thermal conductivities of the air and glass were $0.025Wm^{-1}k^{-1}$ and $0.8Wm^{-1}k^{-1}$ respectively. According to him these properties ensured that even when the outer surface of the window reached 36°C, the temperature on the classroom side would not exceed 20°C, thereby providing a comfortable learning environment.

The physics teacher questioned the validity of both the stated indoor temperature and the quoted thermal conductivity of the glass. The teacher therefore requested one of the S. 6 physics students to investigate the claims using a sample of the same glass material to verify its thermal

Turn over

conductivity and determine whether the classroom side temperature could indeed be maintained at or below 20°C.

In addition, the student was asked to explain how the outer surface of the window could be modified to reduce the amount of energy absorbed from incident light and to describe the operation of a device that could be fitted near the door way to monitor un authorized entry into the house.

Task

You have been selected by the Physics teacher,

(a) Using a data provided, analyse the situation and address the concern expressed by the students about the window quality.

(b) Explain the modification on the window and describe the operation of the installed device.

Part II

Answer one item from this part

Item 5

Several Schools in Uganda experience difficulties in making announcements during assemblies because they lack effective public address system. To solve this problem, the headteacher of a certain secondary school purchased a portable loud speaker system. However, the speaker could only operate for a short period before stopping.

The headteacher suspected that the fault could be due to either a high internal resistance or a high electromotive force (emf) of the battery used to charge the speaker. The battery was used to change the power bank inside the speaker and when fully charged, the power bank was expected to supply energy to the speaker for a longer duration.

An S. 5 student volunteered to investigate whether the battery was responsible for the problem. The student connected the battery in series with a variable resistor and an ammeter. When the resistance has adjusted to 21Ω, the current recorded was 0.48A. When the resistance was increased to 36Ω, the current decreased to 0.30A.

The power bank inside the speaker consisted of the super capacitors, each having a capacitance of 1.0F, connected in series.

The headteacher then asked the student to explain how the capacitors became charged by the battery and determine the time required by the speaker to become fully charged. In addition, the requested an explanation of how devices could be connected to convert the direct current (d.c) supplied by the battery into alternating current (a.c) suitable for operating the schools television set.

The S. 5 student was unable to provide satisfactory answers to these questions and therefore sought assistance from the S. 6 Physics students.

Task

CamScanner

You have been selected, analyse the information given in the scenario and fully address the claims and concern.

✓ **Item 6**

Listening to religious radios is highly advisable for older adults to promote emotional comfort and mental stimulation. Opio visited his parent with his child who was doing Physics in S. 6. The parent complained to Opio of the challenge they had with their radio set which often failed to tune in their favourable religious program. The radio operated at a frequency of 61.2MHz. Opio opened the radio and found this tuning circuit as indicated in figure (i) very week.

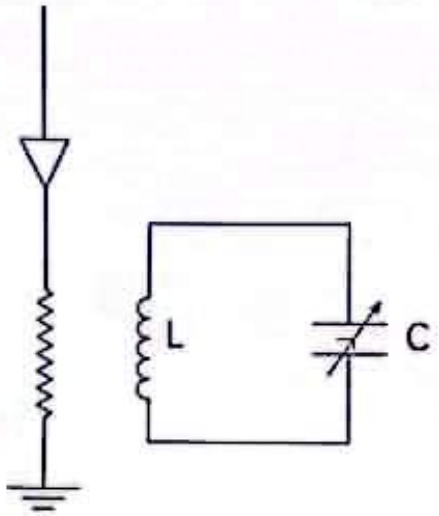


Figure (i) resonance circuit for tuning radio.

Opio decided to install a new stronger variable capacitor and inductor. He purchased materials and made the two smaller capacitors each of plate separation 2mm and maximum area of overlap $5.0 \times 10^{-2} m^2$ filled with a material of dielectric constant 4. The two capacitors were joined in series and connected with a rator for tuning.

He made a pure inductor, L_1 from 10cm air cored solenoid of 100 turns and $1.2 \times 10^{-3} m^{-2}$ cross sectional area Opio requested for his child to perform necessary calculations to establish whether the new circuit can tune to the desired religious channel.

Opio's parent also took advantage of the presence of the grandchild and asked for the explanation of the structure and operation used by the radio charging device being used because, the grandparent had always wondered how the charger can be connected in the socket which was labelled to supply 240V to charge a radio whose input required on 12V.

Support materials.

Permeability of free space $\mu_o = 4\pi \times 10^{-7} Hm^{-1}$

Permeability of free space $\varepsilon_o = 8.85 \times 10^{-12} Fm^{-1}$

Task.

As a S. 6 Physics student analyse the information in the scenario and answer the requests made by Mr. Opio and his parent.

END